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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CST) (CBSC) Degree Examinations

December 2024 - January 2025

22CST123 – COMPUTATIONAL STRUCTURAL MECHANICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing any Two full questions from Unit - I and Unit – II each, and One full question from Unit - III.

Assume any missing data suitably and mention the same.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Define static indeterminacy and kinematic indeterminacy. How are they found in pin jointed frames and axially rigid beams? | 06 | L1 | 1 | 1 |
| b) Mention the characteristics of stiffness matrix. | 04 | L1 | 1 | 1 |
| c) Develop the stiffness matrix for the beam shown in Fig. 1(c) with the given coordinates. | | | | |

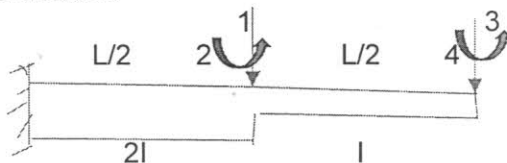


Fig. 1(c)

- | | | | | |
|--|----|----|---|---|
| 2. a) Deduce that stiffness matrix and flexibility matrix are inverse of each other. | 04 | L2 | 1 | 1 |
| b) State and explain: (i) Principle of minimum potential energy, (ii) Principle of minimum complementary energy. | 06 | L2 | 1 | 1 |
| c) Develop the flexibility matrix for the beam shown in Fig. 2(c) with the given coordinates. | | | | |

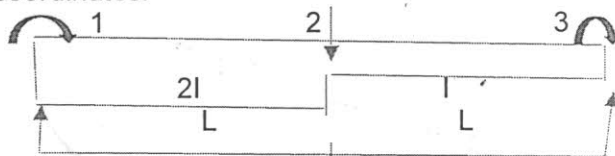


Fig. 2(c)

10	L2	1	1
20	L3	2	2

3. Analyse the rigid joint frame shown in Fig. 3 by flexibility method. Take uniform EI for all members. Treat the reactions at the hinge as redundant.

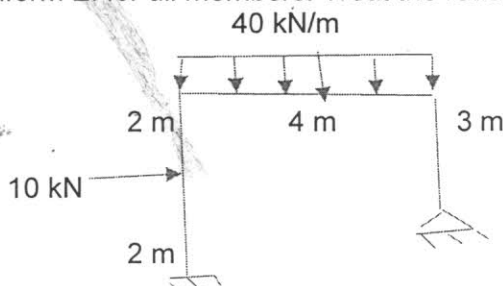


Fig. 3

Unit – II

4. Analyze the beam shown in Fig. 4 by stiffness method using displacement transformation matrix. Draw BMD and SFD. Assume uniform EI for all spans.

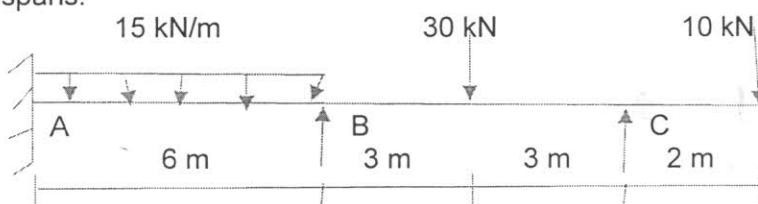


Fig. 4

20	L3	3	2
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P.T.O.

5. Compute the forces in the members of the truss shown in Fig. 5 by stiffness method. Take $(AE/L) = 1$ for all members.

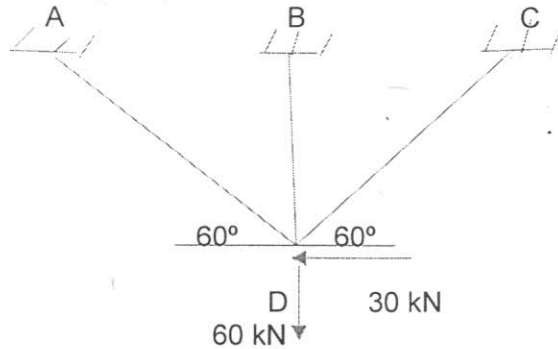


Fig. 5

20 L3 3 2

6. Analyze the beam shown in Fig. 6 by direct stiffness method. Assume constant EI throughout. Sketch the BMD and SFD.

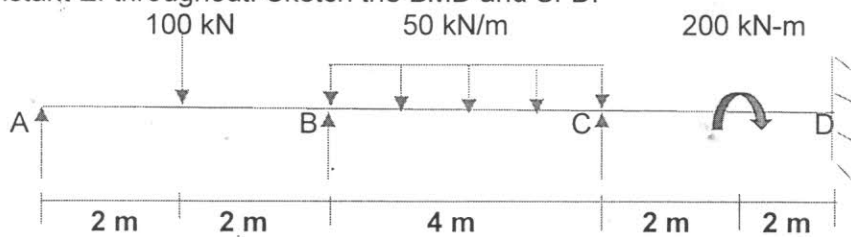


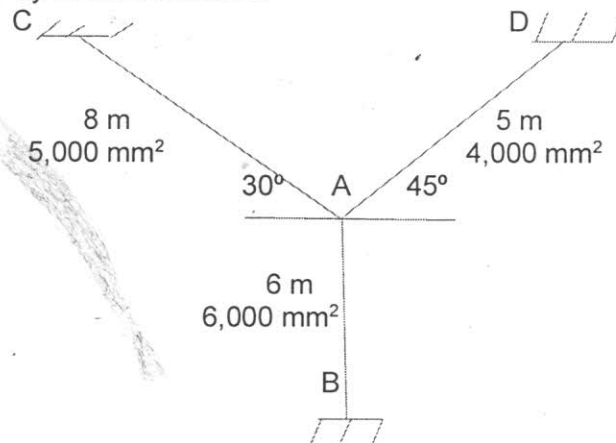
Fig. 6

20 L3 4 2

Unit – III

7. Compute the forces in the members of the pin jointed frame shown in Fig. 7 by stiffness method.

20 L3 3 2



Q. 7 – Fig. 7

All the members are cooled to 20°C .
Assume $E = 200 \text{ GPa}$, $\alpha = 1.2 \times 10^{-5}/^{\circ}\text{C}$.

8. a) Explain:
(i) Skyline storage technique
(ii) Band width minimization
b) Solve the following equations by Gauss Elimination method:
 $x + y + z = 6$, $3x + 3y + 4z = 20$, $2x + y + 3z = 13$

10 L2 5 2

10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
